

Laplace transform:

$\mathcal{L}\{y'\}(\omega) = \omega Y(\omega) - y(0)$
 $\mathcal{L}\{y''\}(\omega) = \omega^2 Y(\omega) - \omega y(0) - y'(0)$
 $\mathcal{L}\{y'''\}(\omega) = \omega^3 Y(\omega) - \omega^2 y(0) - \omega y'(0) - y''(0)$

Convolution theorem:

$\mathcal{L}\{f * g\}(\omega) = \mathcal{L}\left\{\int_0^t f(t-\tau)g(\tau) d\tau\right\} = F(\omega) G(\omega)$

Some transforms:

$\mathcal{L}\{u(t-a)\} = e^{-as} \cdot \frac{1}{s}$
 $\mathcal{L}\{e^{at} \sin(bt)\} = \frac{b}{(s-a)^2 + b^2}$
 $\mathcal{L}\{e^{at} \cos(bt)\} = \frac{(s-a)}{(s-a)^2 + b^2}$
 $\mathcal{L}\{t^n e^{at}\} = \frac{n!}{(s-a)^{n+1}}, n=1,2,3,\dots$

$\mathcal{L}\{e^{at} f(t)\} = F(s-a)$

Interpolation:

Lagrange: $L_j = \prod_{m=0, m \neq j}^n \frac{x - x_m}{x_j - x_m}$
Newton: $L(x) = \sum_{j=0}^n L_j y_j$
 $x_0, y_0 = [y_0]$
 $x_1, y_1 = [y_1]$
 $x_2, y_2 = [y_2]$
 $x_3, y_3 = [y_3]$
 $x_4, y_4 = [y_4]$
 $[y_0, y_1] = \frac{y_1 - y_0}{x_1 - x_0}$
 $[y_0, y_1, y_2] = \frac{[y_1, y_2] - [y_0, y_1]}{x_2 - x_0}$
 $N(x) = [y_0] + [y_0, y_1](x-x_0) + [y_0, y_1, y_2](x-x_0)(x-x_1) + \dots$

Fourier Analyse:

Periode $T = 2L$
 $f(x) = \sum_{n=-\infty}^{\infty} C_n \exp(i \frac{n\pi x}{L})$
 $f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(\frac{n\pi x}{L}) + b_n \sin(\frac{n\pi x}{L})]$
 $C_n = \begin{cases} \frac{1}{2} (a_n - ib_n), n > 0 \\ \frac{1}{2} (a_n + ib_n), n < 0 \end{cases}$
 $a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx = C_0$
 $a_n = \frac{1}{L} \int_{-L}^L f(x) \cos(\frac{n\pi x}{L}) dx$
 $b_n = \frac{1}{L} \int_{-L}^L f(x) \sin(\frac{n\pi x}{L}) dx$
 $C_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-i \frac{n\pi x}{L}} dx$

Funksjoner:

$f(-x) = -f(x)$ (odd)
 $f(-x) = f(x)$ (even)
Odd extension $\rightarrow B_n$
Even extension $\rightarrow A_0, A_n$
Non-periodic functions Fourier extensions:
Ex: $f(x) = \sin(x)$
Ex: $g(x) = \cos(x), |x|$
Even: $A_n = \frac{1}{L} \int_0^L f(x) dx$
Odd: $B_n = \frac{1}{L} \int_0^L f(x) \sin(\frac{n\pi x}{L}) dx$

Fourier transform:

$\mathcal{F}\{f * g\}(\omega) = \sqrt{2\pi} [\mathcal{F}\{f\} \cdot \mathcal{F}\{g\}]$
 $\mathcal{F}\{u'\}(\omega) = i\omega \mathcal{F}\{u\}(\omega)$
 $\mathcal{F}\{u''\}(\omega) = -\omega^2 \mathcal{F}\{u\}(\omega)$

PDE (FT):

Ex: $u_t = u_x + u, u(x,0) = \begin{cases} \frac{\sin(x)}{x}, x \neq 0 \\ 1, x = 0 \end{cases}$
- Fourier transformer begge sider mhp. x. ($\hat{u}(\omega, t)$)
- Gjør om til ODE ved bruk av $\mathcal{F}\{u_t\} = \frac{d\hat{u}(\omega, t)}{dt} = \hat{u}(\omega, t)(1+i\omega)$
- Løs ODE. ($\hat{u}(\omega, t) = C(\omega)e^{(1+i\omega)t}$)
- Transformer initial bet. for $C(\omega)$. ($C(\omega) = \hat{u}(\omega, 0) = \begin{cases} \sqrt{\pi/2}, |\omega| < 1 \\ 0, |\omega| > 1 \end{cases}$)
- Sett inn. ($\hat{u}(\omega, t) = \begin{cases} \sqrt{\pi/2} e^{(1+i\omega)t}, |\omega| < 1 \\ 0, |\omega| > 1 \end{cases}$) og invers transformer

Numersk integrasjon:

$\int_a^b f(x) dx \approx \sum_{j=0}^n f(x_j) \omega_j = Q[f]$, uløst fvs: $\int_a^b P_n(x) dx = \sum_{j=0}^n f(x_j) \int_a^b \omega_j(x) dx = \sum_{j=0}^n f(x_j) \omega_j$
Fra a til b. $Q[f](a,b)$
Midtpunkt: $QM[f] = (b-a) f(\frac{a+b}{2})$
Trapezmetode: $QT[f] = (b-a) (\frac{1}{2} f(a) + \frac{1}{2} f(b))$
Simpsons metode: $QS[f] = \frac{b-a}{6} (f(a) + 4f(\frac{a+b}{2}) + f(b))$
Simpsons metode krever at $b-a$ er et partall-multiple av h . Dvs. $h < \epsilon$, h er størst $\frac{b-a}{n} < \epsilon$, $n=2,4,\dots$
Error: $E_{QM} = \frac{(b-a)^2}{6} \max_{x \in [a,b]} |f''(x)|$
 $E_{QT} = \frac{(b-a)^3}{12} \max_{x \in [a,b]} |f''(x)|$
 $E_{QS} = \frac{(b-a)^4}{180} \max_{x \in [a,b]} |f^{(4)}(x)|$
Precision d: $\int_a^b x^d dx = Q[x^d](a,b)$ for $k=1,2,\dots,d$

Parseval identitet:

$\int_{-L}^L f(x) dx$ distribuer og at $f(x) = \sum_{n=-\infty}^{\infty} C_n e^{i \frac{n\pi x}{L}}$
 $\frac{1}{2L} \int_{-L}^L f(x)^2 dx = \sum_{n=-\infty}^{\infty} |C_n|^2 = a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$
 $E_N = \int_{-L}^L (f(x) - F_N(x))^2 dx = \int_{-L}^L f(x)^2 dx - L(2a_0^2 + \sum_{n=1}^N (a_n^2 + b_n^2))$ (Fourier Error)

D'Alembert:

Bølge ligning: $u_{tt} = c^2 u_{xx}$
Init.: $u(x,0) = f(x)$
 $u_t(x,0) = g(x)$
 $u(x,t) = \frac{1}{2} [f(x+ct) + f(x-ct)] + \frac{1}{c} \int_{x-ct}^{x+ct} g(\tau) d\tau$

Rounding errors:

Ex: $\epsilon = 10^{-16}$, $O(h^2) = O(\frac{\epsilon}{h^2}) \Rightarrow h^4 = O(\epsilon) \Rightarrow h = O(\epsilon^{1/4}) = O(10^{-4})$

Newtons metode:

$X_{k+1} = X_k - \frac{f(X_k)}{f'(X_k)}$
- Funksjonen må være glatt slik at den deriverte oppfører seg normalt i nærheten av roten og ikke er lik null for å unngå å dele på null.
Kvadratisk konvergens, $P=2$:
 $|X_{k+1} - r| \leq M |X_k - r|^2$
(Avhengig av oppgaven)

Bisection metoden:

- To punkter a, b slik at $f(a)f(b) < 0$
- Finn så midtpunktet $c = \frac{a+b}{2}$
- Sjekk om $f(c)=0$, hvis ja er roten funnet.
Hvis ikke, fortsetter metoden.
- $f(a)f(c) < 0$ er roten på intervallet (a,c)
- Sett derfor $a_1 = a, b_1 = c, c_1 = \frac{a_1+b_1}{2}$
- $f(a)f(c) > 0 \Rightarrow$ roten $\in (c,b)$
Iterations $k \gg \frac{\log(\frac{b-a}{2 \cdot \text{tol}})}{\log(2)}$

Fixed point: $x = g(x)$

Anta at $g(x)$ er kontinuert på intervallet $[a,b]$. Anta også at $g'(x)$ eksisterer og at det finnes en konstant $L < 1$.
(1) $a < g(x) < b \forall x \in [a,b]$
(2) $|g'(x)| \leq L < 1 \forall x \in [a,b]$
Hvis (1) holder, eksisterer det minst et fikspunkt på intervallet $[a,b]$
Hvis (1) og (2) holder, er dette fikspunktet unikt og iterasjonen vil konvergere til dette fikspunktet for alle $x_0 \in [a,b]$
Error: $|x_{k+1} - r| \leq \frac{1}{1-L} |g(x_k) - x_k|$
 $E < \text{tol}$: $k \gg \frac{\ln((1-L) \cdot \text{tol})}{\ln(L)} - 1$

FDM: (Forward/backward gir ikke høyere pres. enn 10^{-16})

Forward: $u'(x) = \frac{u(x+h) - u(x)}{h} + O(h)$
Backward: $u'(x) = \frac{u(x) - u(x-h)}{h} + O(h)$
Central: $u'(x) = \frac{u(x+h) - u(x-h)}{2h} + O(h^2)$
 $u''(x) = \frac{u(x+h) - 2u(x) + u(x-h))}{h^2} + O(h^2)$

Forward Euler (Varme):

(Eksplicit)
 $u_t = u_x + u, u(x,0) = f(x), u(0,t) = u(L,t) = C$
 $u_k = \frac{u(x, t+k) - u(x, t)}{k} + O(k)$
 $u_{xx} = \frac{u(x+h, t) - 2u(x, t) + u(x-h, t))}{h^2} + O(h^2)$
Scheme: $\frac{u_{m+1}^n - u_m^n}{k} = \frac{u_{m+1}^n - 2u_m^n + u_{m-1}^n}{h^2}$

Fleere dimensjoner:

$f_n(x_1, x_2, \dots, x_n)$
 $J_{i,j} = \begin{bmatrix} \frac{\partial f_i}{\partial x_1} & \dots & \frac{\partial f_i}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}$
 $X_{k+1} = X_k - J^{-1} f(X_k)$
- Hvor J er jacobimatriksen og x_k er en $1 \times n$ vektor.
 $\Rightarrow J \Delta K = -f(X_k), \Delta K = X_{k+1} - X_k$
1) Finn J og $f(X_k)$
2) Løs for ΔK og sett inn i $X_{k+1} = X_k + \Delta K$

Taylor ekspansjon:

$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$
 $u(x-h) = \sum_{n=0}^{K-1} \frac{u^{(n)}(x)}{n!} (-h)^n + O(h^K)$

Konvergensrate:

$\rho = \lim_{k \rightarrow \infty} \frac{\|e_{k+1}\|}{\|e_k\|}$
Konvergensrate ρ hvis: $e_{k+1} \leq C e_k^\rho, C > 0$
Halverer steglengden gir: $\rho = \lim_{k \rightarrow \infty} \frac{\|e_k\|}{\|e_{k-1}\|}$

Differensialligninger:

$$ay'' + by' + cy = 0 \Rightarrow ar^2 + br + c = 0$$

* En løsning: $y = (A + Bx)e^{rx}$

* To løsninger: $y = Ae^{r_1x} + Be^{r_2x}$

* Ingen løsning: $r = \Delta \pm \pm i$

$$y = e^{\Delta x} (A \sin(\pm x) + B \cos(\pm x))$$

PDEs:

Lineær $\rightarrow$ alle u og dens deriverte er av grad 1.

Homogen $\rightarrow$ alle ledd avhenger av u

Separation of variables:

1) Sett $u(x,t) = F(x)G(t)$ og finn ODE for F og G .

2) Finn alle ikke-triviale løsninger $u(x,t) = \sum_{n=1}^{\infty} F_n(x)G_n(t)$ (Superposisjon)

3) Finn alle ko-eff. som tilfredsstiller initial betingelsen.

Runge-Kutta metoder:

A Runge-Kutta method with Butcher tableau (I) to solve the ODE $y' = f(t, y)$ is defined as (II)

$$\begin{array}{c|ccc} I & S=1 & C_1 & a_{11} & \dots & a_{1s} \\ & \vdots & \vdots & \vdots & \ddots & \vdots \\ & S=s & C_s & a_{s1} & \dots & a_{ss} \\ \hline & & & b_1 & \dots & b_s \end{array} \quad II \quad \begin{array}{l} k_1 := f(t_n + c_1 h, y_n + h \sum_{j=1}^s a_{1j} k_j) \\ \vdots \\ k_s := f(t_n + c_s h, y_n + h \sum_{j=1}^s a_{sj} k_j) \\ y_{n+1} := y_n + h \sum_{j=1}^s b_j k_j \end{array}$$

Eulers metode:

$$y' = \frac{dy}{dx} = f(x_n, y(x_n)) \mid y(x_0) = y_0$$

$$y(x_n + h) \approx y(x_n) + h y'(x_n)$$

$$y_{n+1} = y_n + h f(x_n, y_n)$$

Velg ny steglengde: (RR)

$$h_{\text{new}} = P \left(\frac{\text{Faktor}}{1 + \text{Faktor}} \right)^{\frac{1}{P}} \cdot h_{\text{old}}$$

$P=0.5$ (Pessimist factor)

Nyttig:

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\int u \cdot v' = u \cdot v - \int u' \cdot v$$

$$\int_{-\pi}^{\pi} e^{i(n-m)x} dx = \begin{cases} 2\pi, & n=m \\ 0, & n \neq m \end{cases}$$

$$u(x-a) = \begin{cases} 1, & x > a \\ 0, & x < a \end{cases}$$

$$\delta(x-a) = \begin{cases} \infty, & x=a \\ 0, & x \neq a \end{cases}$$

$$\sin\left(\frac{n\pi}{2}\right) = \sin\left(\frac{(2k+1)\pi}{2}\right) = (-1)^k \Rightarrow \frac{\sin\left(\frac{n\pi}{2}\right)}{n\pi} = \frac{(-1)^k}{(2k+1)\pi}$$

$$\cos\left(\frac{n\pi}{2}\right) = \cos\left(\frac{(2k)\pi}{2}\right) = (-1)^k \Rightarrow \frac{\cos\left(\frac{n\pi}{2}\right)}{n\pi} = \frac{(-1)^k}{(2k)\pi}$$

$$\int_{-\infty}^{\infty} \delta(x-a) \delta(x-a) = \delta(a)$$

$$\sin\left(\frac{(2k+1)\pi}{2}\right) = \begin{cases} 1, & k=0,2,4 \\ -1, & k=1,3,5 \end{cases}$$

$$f(x) + \delta(x-a) = f(x-a)$$

Heuns metode:

Kan også skrives:

$$u_{n+1} = y_n + h f(x_n, y_n) \quad (\text{Euler})$$

$$k_1 = f(x_n, y_n)$$

$$k_2 = f(x_n + h, y_n + h k_1)$$

$$y_{n+1} = y_n + \frac{h}{2} (f(x_n, y_n) + f(x_{n+1}, u_{n+1}))$$

$$y_{n+1} = y_n + \frac{h}{2} (k_1 + k_2)$$

Diskret FT:

$$\hat{F}_N^{-1} = \frac{1}{N} \hat{F}_N$$

$$\hat{F}_N = \left(e^{-2\pi i j k / N} \right)_{j,k=0}^{N-1}, \quad \hat{f} = (f_0, f_1, \dots, f_{N-1}), \quad f_j = F\left(\frac{2\pi j}{N}\right)$$

$$\hat{f} = \hat{F}_N^{-1} \hat{f} \mid \hat{f} = \left(\frac{1}{2}, \frac{1}{2}, 0\right), \quad \hat{F}_N = \left(e^{-2\pi i j k / 4} \right)_{j,k=0}^3$$

$$\begin{array}{c|cccc} k \backslash j & 0 & 1 & 2 & 3 \\ \hline 0 & e^0 & e^0 & e^0 & e^0 \\ 1 & e^0 & e^{-\pi i} & e^0 & e^{-\pi i} \\ 2 & e^0 & e^{-\pi i} & e^0 & e^{-\pi i} \\ 3 & e^0 & e^{-\pi i} & e^0 & e^{-\pi i} \end{array}$$

$$\hat{f} = \hat{F}_N^{-1} \hat{f} = \begin{pmatrix} 2 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

$$k_1 = f(x_n, y_n)$$

$$k_2 = f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} k_1\right)$$

$$y_{n+1} = y_n + h k_2$$

Runge Kutta eks:

a) Utfør én iterasjon med steglengde $h=0.1$ ved å bruke metoden ovenfor på problemet:

$$y' = y + x y^2, \quad y(1) = 1.0$$

$$y_2' = y_1 y_2, \quad y_2(1) = -1.0$$

$$\vec{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \quad \vec{f} = \begin{bmatrix} y_1 + x y_1^2 \\ y_1 y_2 \end{bmatrix}, \quad \text{Initial values: } \vec{y}(1) = \vec{y}_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad x_0 = 1$$

For $n=0$ we get:

$$\vec{k}_1 = [2, -1]^T$$

$$\vec{k}_2 = f\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2} \vec{k}_1\right) \quad (\text{oppdater } x \text{ en! } (x_0 + \frac{h}{2}))$$

$$= f(1.05, [1.1, -1.05]^T)$$

$$= [2.22576, -1.155]^T$$

$$y_1 = [1.2258, -1.116]^T$$

b) Find the stability function $R(z)$ for this method. Find also the corresponding stability interval

Use the method to solve the linear test equation $y' = \lambda y, \quad \lambda < 0$

The stability function $R(z)$ is defined by $y_{n+1} = R(z) y_n, \quad z = h \lambda$

which in this case can be found by:

$$k_1 = \lambda y_n$$

$$k_2 = \lambda \left(y_n + \frac{h}{2} \lambda y_n \right) = \lambda \left(1 + \frac{h}{2} \lambda \right) y_n$$

$$y_{n+1} = y_n + h \left(1 + \frac{h}{2} \lambda \right) y_n$$

$$= y_n \left(1 + z + \frac{z^2}{2} \right)$$

$$R(z) = 1 + z + \frac{z^2}{2}$$

Stability interval $S = \{z \in \mathbb{R} : |R(z)| \leq 1\}$

$$1 + z + \frac{z^2}{2} \leq 1 \quad 1 + z + \frac{z^2}{2} \geq -1$$

$$\Leftrightarrow 0 \leq z \leq 2 \quad \Leftrightarrow z \in \mathbb{R}$$

$$S = [-2, 0]$$

Runge-Kutta eks:

$$C_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad y'(t) = -6y(t) = f(t, y(t))$$

$$C_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad y(0) = 1$$

$$k_1 = f(x_n, y_n) = -6y_n$$

$$k_2 = f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} k_1\right)$$

$$k_2 = f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} (-6y_n)\right)$$

$$= f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} (-6y_n)\right)$$

$$= -6\left(y_n - 3hy_n\right) = -6y_n(1 - 3h)$$

$$y_{n+1} = y_n + h \sum_{j=1}^3 b_j k_j$$

$$= y_n + h \left(\frac{1}{4} k_1 + \frac{3}{4} k_2 \right)$$

$$= y_n + \frac{h}{4} (-6y_n) + 3 \left(-6y_n - 4hy_n \right)$$

$$= y_n - \frac{3}{2} h y_n - \frac{18}{4} h y_n + \frac{3}{4} h^2 y_n$$

$$= y_n - 6y_n h + 18h^2 y_n$$

$$= y_n (1 - 6h + 18h^2)$$

$$\Rightarrow R(h) = 1 - 6h + 18h^2$$

Stability means that y_n does not tend to infinity for $n \rightarrow \infty$. Since $y_n = [R(h)]^n y(0)$ and $R(h) = 1 - 6h + 18h^2 > 0 \quad \forall n \in \mathbb{R}$ all we need to guarantee is $R(h) \leq 1$. $1 - 6h + 18h^2 \leq 1 \Leftrightarrow 18h^2 - 6h \leq 0 \Leftrightarrow 6h(3h - 1) \leq 0 \Leftrightarrow 3h - 1 \leq 0 \Leftrightarrow h \leq \frac{1}{3}$

FT eks:

$$F(f) = (wF(f)) \Rightarrow F(f'') = -w^2 F(f)$$

$$F(x) = x^2 e^{-x^2}$$

$$g(x) = e^{-x^2} \Rightarrow g'(x) = -2x e^{-x^2} \Rightarrow g''(x) = 4x^2 e^{-x^2} - 2e^{-x^2}$$

$$g''(x) = 4x^2 e^{-x^2} - 2e^{-x^2} \Rightarrow F(x) = \frac{1}{4} g''(x) + \frac{1}{2} e^{-x^2} = \frac{1}{4} g''(x) + \frac{1}{2} e^{-x^2}$$

$$F(f) = \frac{1}{4} F(g'') + \frac{1}{2} F(e^{-x^2})$$

$$= -\frac{1}{4} w^2 F(e^{-x^2}) + \frac{1}{2} F(e^{-x^2}), \quad F(e^{-x^2}) = \frac{1}{\sqrt{2}} e^{-\frac{x^2}{2}}$$

$$= e^{-\frac{x^2}{2}} \left(\frac{1}{4\sqrt{2}} - \frac{w^2}{4\sqrt{2}} \right)$$

Fundamental period:

for 2π -periodisk. $f(x+2\pi) = f(x)$

$$g_1(x) := f(a-x), \quad g_2(x) := f(x+a), \quad g_3(x) := f(x^*), \quad g_4(x) := a+a - (f(\frac{x}{2}+a))^*$$

$$1. \quad g_1(x + \frac{2\pi}{a}) = f(a - (x + \frac{2\pi}{a})) = f(a - x) = g_1(x) \Rightarrow T = \frac{2\pi}{a}$$

$$2. \quad g_2(x + 2\pi) = f(x + 2\pi + a) = f(x + a) = g_2(x) \Rightarrow T = 2\pi$$

$$3. \quad \text{Not periodic} \quad 4. \quad T = 2\pi a$$

- F er en funks med periode P , da er $f(x)$ periodisk med $\frac{P}{|a|}$.

Crank-Nicolson:

$$(2+2r)u_{i+1}^{j+1} - r(u_{i+1}^{j+1} + u_{i-1}^{j+1})$$

$$= (2-2r)u_i^{j+1} + \frac{1}{2}(u_{i+1}^j + u_{i-1}^j)$$

$$\text{der } r = \frac{\Delta t}{\Delta x^2}$$

Numerisk ODE: Skriv om til et ligningssystem og approx. $y(0.1)$.

$$y'' - 3y' + 2y = 0 \mid y(0) = 0, \quad y'(0) = 1$$

Eulers metode:

$$x(t) = y'(t) \Rightarrow \vec{y} = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} 3x - 2y \\ x \end{pmatrix}$$

Velg $h=0.1$ og definer $x_n = x(nh)$ og $y_n = y(nh)$

$$\begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = \begin{pmatrix} x_n \\ y_n \end{pmatrix} + h \begin{pmatrix} 3x_n - 2y_n \\ x_n \end{pmatrix} \mid \begin{matrix} x_0 = x(0) = y'(0) = 0 \\ y_0 = y(0) = 1 \end{matrix} \Rightarrow \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$n=0$:

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + 0.1 \begin{pmatrix} 3 \cdot 0 - 2 \cdot 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -0.2 \\ 1 \end{pmatrix}$$

$$y(0.1) = y_1 = 1$$

ODEs and other reduction

An m -th-order initial value problem $u^{(m)} = f(t, u, u', \dots, u^{(m-1)})$

$$u(t_0) = u_0, \dots, u^{(m-1)}(t_0) = u_0^{(m-1)}$$

can be written as a first-order system of ODEs

$$y_1' = y_2$$

$$\vdots$$

$$y_{m-1}' = y_m$$

$$y_m' = f(t, y_1, y_2, \dots, y_{m-1}, y_m)$$

$$y, (t_0) = u_0 \quad \text{with } y = u, y' = u', \dots, y^{(m-1)} = u^{(m-1)}$$

$$y_{m-1}(t_0) = u_0^{(m-2)}$$

$$y_m(t_0) = u_0^{(m-1)}$$